



ExcellentNYC

Deliverable 2

Project Proposal & Preliminary Design

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Company Name: Harborview Services

University at Buffalo

Course Number: MGS 102

Project plan / Work breakdown structure:

MGG 102

Deliverable 3 - Tasks

- [Main] IS Reports (Not Started, Main) Apr 24
- [Main] Input Forms (Not Started, Main) Apr 24
- [Main] IS Tables (Not Started, Main) Apr 24
- [Main] Product Prototype Writeup (Not Started, Main) Apr 24
- [Main] QA & Test (Not Started, Need review to final, Main)

Shi Wei

- [IS Tables] Configure Tables (Input masks, rules) (Not Started, Pre Req) Apr 24
- [IS Tables] Build Tables (Not Started, Pre Req) Apr 24
- [IS Tables] Create & Fill DB with Real Data (Not Started, Pre Req) Apr 24

Rob

- [IS Reports] Buildout Report Design (Not Started, Pre Req) Apr 24
- [IS Reports] Write Report Outline (What/How many?) (Not Started, Pre Req) Apr 24

Abdullah

- [Report/Forms] Implement Business logic (Flow Chart) (Not Started, Pre Req) Apr 24
- [IS Tables] Build Tables (Not Started, Pre Req) Apr 24
- [IS Tables] Configure Tables (Input masks, rules) (Not Started, Pre Req) Apr 24

Amir

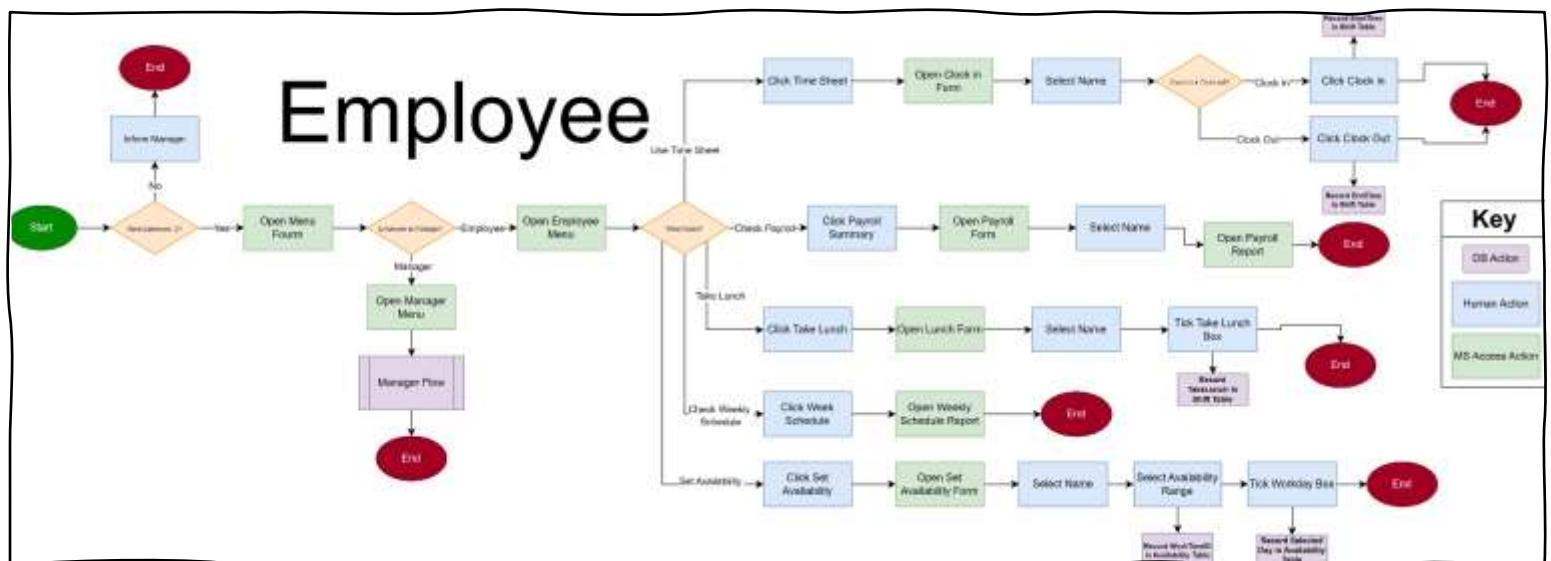
- [Write-up] Write out Product Prototype (Not Started, Pre Req) Apr 24
- [QA & Test] Test & Enforce Business Rules (Not Started, Pre Req) Apr 24

Belinda

- [Input Forms] Write Form Outline (What/How many?) (Not Started, Pre Req) Apr 24
- [Input Forms] Buildout Form Design (Not Started, Pre Req) Apr 24

Flowchart Illustrating Workflow:

There have been multiple iterations of the flowchart. From extremely specific steps that are redundant to outright unnecessary designs. The final design focuses only on the employee side of the workflow. The purpose of the diagram is to show system logic, which will be used when building out the reports/forms. The blue boxes represent user actions, green boxes are Microsoft actions, and pink boxes are functions done by the database. The user first must have their information in the system, hence the first check. If they do not, they must talk to their manager and get enrolled in the system. If it is an established employee, then they can proceed. Users can clock in, clock out, take lunch, check schedules, and set or change availability. Dividing and outlining each branch makes it easy to segment each action when building the information system. Each action results in an interaction of the information system, from recording data to returning queries. Each branch represents/aligns with our business logic. For instance, users can click "clock out" without clocking in. This means when employees make a mistake like forgetting to clock in, they can mention it to managers so they can edit and change it in the back end via additional forms.



UI Mockups of Forms:

The focus was on making the system easy to use while keeping the data clean. A “Role Selection” screen was used at the start so that employees and managers only see the forms they need, which keeps the interface from getting cluttered and helps with security. Large buttons were added to the Role Selection to improve accessibility and ease of selection. On the data entry side, dropdown menus and data pickers were used for things like the “Department” and “Shift” fields. This is a way it forces users to enter the right information, so the database does not end up formatting errors. To handle the one-to-many relationships in the system, the manager view uses a "subform" layout. This allows a manager to select one department and instantly see a list of many employees or shifts assigned to it without leaving the page. Finally, the use of large, high-contrast buttons and a simple layout ensure the portal works well on a tablet or a quick-service kiosk where people need to clock in and out fast without searching for tiny links.



03/10/2026 , 08:00:00 AM

Time Clock

Schedule for today:

Monday Shift: 8:00 AM - 2:00 PM

Take Lunch

☒ Yes ☒ No

Employee ID

 000111226

CLOCK-IN

08:00:00 AM

CLOCK-OUT

02:13:06 PM

< Prev Next >

Dropdown selection, reduces data entry errors

High- Contrast buttons improve visibility for users

Department filter Sales

Search

Department Table

	ID	Department
➤	01	Sales
	02	Finance
	03	Marketing

Details

	ID	Full Name	Position	Hourly Rate
➤	000111226	Matt Smith	Director	\$ 45.00
	000111227	Linda Miller	Project Manager	\$ 34.40
	000111228	Hudson Williams	Administrator	\$ 29.00
	000111229	Ilya Johnson	Team Leader	\$ 21.23

< Prev Next >

Adding search and filter bars directly solves the Scheduling Conflict issue by letting managers instantly find and verify individual employee shifts without scrolling.

Use of a unified table structure to prevent clutter

Payroll Summary

Calendar pickers eliminate bad data by forcing a standard format

Payroll Period Start Date 

Payroll Period End Date 

Total Hours Worked 

Total Payroll Cost 

These buttons turn data into a report in seconds

Generate Payroll Report **Export to PDF**

[< Prev](#)

UI Mockups of Reports:

When making these reports, the vision was to make sure they looked very clear and easy to understand for the managers. These reports are used to look at employee data without needing to cross-reference any information or do manual calculations. Each report is built using the ERD diagram and the flowchart, so every field is traced back to a specific source, meaning nothing is entered by hand. This keeps the data reliable and reduces the errors ExcellentNYC was experiencing beforehand. Every report also has a filtering option that lets the managers sort the information by specific filtrations. On the bottom of the records, it shows how many records are being shown and what page the record is on. As a group, we chose to keep our UI yellow and cream colors.

Instead of merging Schedule and Shift information, the Weekly Timesheet Report divides them into distinct columns. This eliminates the need for managers to check individual records in order to identify late arrivals, early departures, or missing clock-ins by providing them with a planned vs. actual comparison. There is also a section on the bottom that calculates the weekly total hours from that page of the report.

From shift clock data, including the HrlyRate column in the Employee table, the Payroll Summary Report computes Regular Pay, OT Pay, and Total Pay.

The Employee Availability Report sorts the WorkTimeID foreign key into something that is easier to read from the WorkTimeOption table, so users can see something like “9 AM – 5 PM” instead of a raw ID number.

The Department Staffing Overview shows the overall headcount, scheduled coverage, and Schedule Coverage % per department using the DepartmentID foreign key that links the Department and Employee tables. The Schedule Coverage % column will flag any department below 80% coverage in red, letting managers know where there are staffing gaps. There is also a section on the bottom that calculates the total headcount from that page of the report.

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Report Mockups — Deliverable 2

Employee Availability Report

Filters:

Employee ID	Employee Name	Dept	Period Start	Period End	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Work Time Option

Records shown: 6 of 38 employees - Page 1 of 7

Accessibility & UX Callouts

Note-Sun Are Yes/No Fields in Availability Table
 Each day column maps directly to a Yes/No field in the Availability table. A blank cell means the employee left that day unscheduled, "MIS", not "No". The report distinguishes these two states so schedulers know whether an employee is unavailable vs. simply did not submit.

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Report Mockups — Deliverable 2

Department Staffing Overview

Filters:

Department	Manager	Total Employees	Scheduled This Week	Present Today	Avg Hrs / Week	Staff Experience	Schedule Coverage %

Records shown: 4 of 4 departments - Page 1 of 1

Total Headcount:

Accessibility & UX Callouts

Staff Experience Column
 The Staff Experience column calculates the time since HireDate across all employees in the department, giving managers insight into the staff.

Entity-Relationship Diagram (ERD):

The Entity-Relationship Diagram (ERD) establishes the layout of data pieces, and shapes how everything connects within the application. Instead of listing them separately, think of each part; like the Department or Employee as a building block linked meaningfully to others. These links mirror actual workplace routines, showing who belongs where, when people are free, and which shift they can take. Through careful setup, one piece leads naturally to another, schedules are attached to shifts, workers are tied into teams, choice about hours are matched up with personal availability, keeping everything organized and clean for the users. Therefore, pulling data later-on will become quicker and smoother, especially when forms are sent directly to store records. Behind the entire structure, early planning is focused on what is needed before it is turned into labeled boxes and lines. Each label gets broken down further until fields make sense alongside their parent groups. Through multiple different versions, unnecessary details were removed to keep the important ones in view and ensure daily use is functional. The design was built to operate smoothly and stay simple.

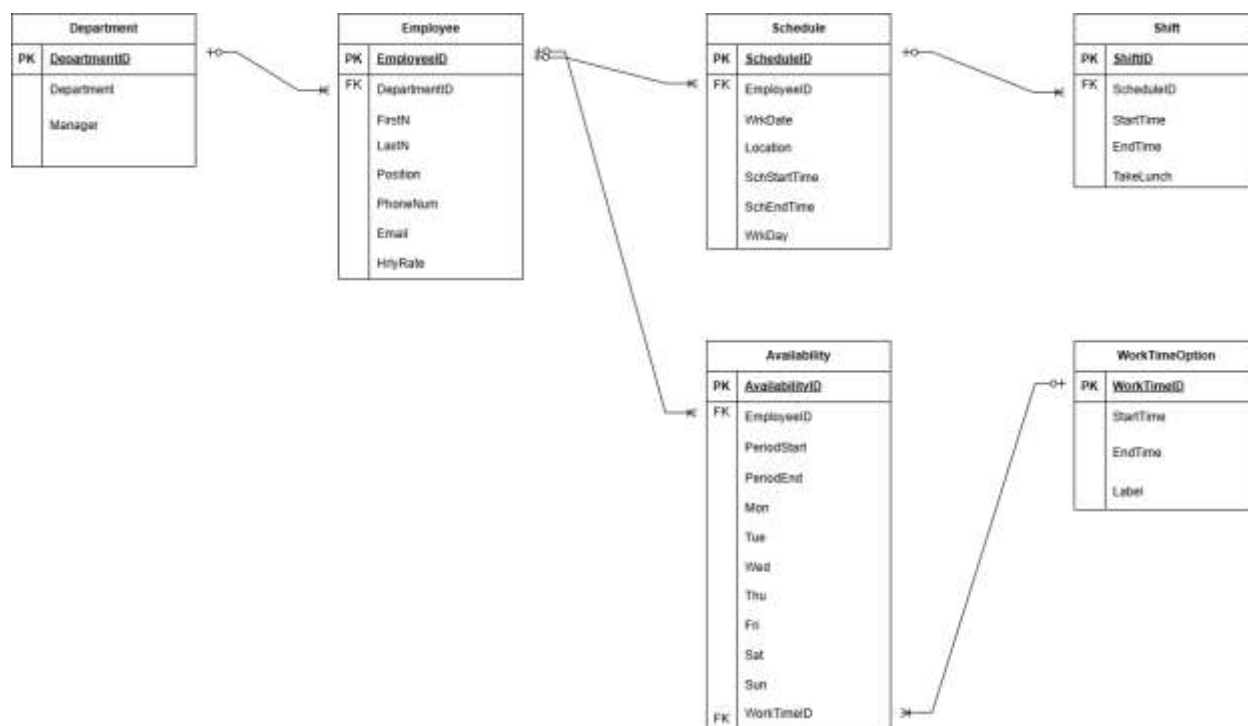


Table Overview:

Employee Table stores all essential information about each employee in the company. It serves as the central table that connects to other tables such as Schedule, Department, and Availability.

Department Table- defines the different departments within the organization and helps categorize employees.

Schedule Table- stores planned work schedules for employees, including when and where they are expected to work.

Shift Table- records actual work activity, specifically when employees clock in and out.

Availability Table- captures when employees are available to work over a defined period. This helps managers create schedules based on employee preferences.

Worktime Option Table- defines standard time ranges that employees can select when indicating availability and, in the future, makes it easier to add new timeslots.

Employee Table

Field Name		Data Type	Description
EmployeeID	Primary Key	Auto Number	Connect to the Schedule table as a foreign key to ensure each employee has a schedule
DepartmentID	Foreign Key	Number	Connect to Department Table to ensure all the employee are in a department
FirstN		Short Text	First name of Employee
LastN		Short Text	Last name of Employee
Position		Short Text	Employee's role in the company
PhoneNum		Short Text	Use to reach the Employee in case of an emergency
Email		Short Text	Company email to reach the Employee
HrlyRate		Numbers	Employees pay rate

Department Table

Field Name		Data Type	Description
DepartmentID	Primary Key	Auto Number	Connect to the Employee table as foreign key

Department	Short Text	Name of the department
Manager	Short Text	Who is the manager of this department

Schedule Table

Field Name		Data Type	Description
ScheduleID	Primary Key	Auto Number	Connect to the Shift table as a foreign key
EmployeeID	Foreign Key	Number	Connect to the employee table to ensure employees have a schedule
WrkDate		Date/Time	The date the employee works
WrkDay		Short Text	The day the employee works
Location		Short Text	Where are the employees working at
SchStartTime		Date/Time	The actual start time or the time the employee needs to clock in by
SchEndTime		Date/Time	The actual end time or the time the employee needs to clock out by

Shift Table

Field Name		Data Type	Description
ShiftID	Primary Key	Auto Number	Unique Shift ID
ScheduleID	Foreign Key	Number	Connect to schedule Table to ensure all employees have a schedule

StartTime	Data/Time	When the employee clock-in
EndTime	Data/Time	When the employee clock-out
TakeLunch	Yes/No	The lunch break that the employee takes

WorkTimeOption Table

Field Name		Data Type	Description
WorkTimeID	Primary Key	Auto Number	Primary Key for WorkTimeOption and connect to availability table as foreign key
StartTime		Data/Time	Start time for the option the employee picks
EndTime		Data/Time	Start end for the option the employee picks
Label		Short Text	Use to display the Start/End time

Availability Table

Field Name		Data Type	Description
AvailabilityID	Primary Key	Auto Number	Primary key for a period
EmployeeID	Foreign Key	Number	Connect to the employee table to ensure all employees can choose availability
PeriodStart		Data/Time	The start date of the period

PeriodEnd	Data/Time	The end date of the period
Mon	Yes/No	Checkbox for employees to indicate Monday availability
Tue	Yes/No	Checkbox for employees to indicate Tuesday availability
Wed	Yes/No	Checkbox for employees to indicate Wednesday availability
Thu	Yes/No	Checkbox for employees to indicate Thursday availability
Fri	Yes/No	Checkbox for employees to indicate Friday availability
Sat	Yes/No	Checkbox for employees to indicate Saturday availability
Sun	Yes/No	Checkbox for employees to indicate Sunday availability
WorkTimeID	Foreign Key	Short Text
		Selection of work time option